ASSIGNMENT 6 — QUESTION 1

Q1 - The Startup Morning: Async Coffee Maker

JavaScript (ES6+) solution with comments

```
// Q1: Async Coffee Maker
// Each step returns a Promise and takes a random 1-2 seconds.

function delay() {
    return 1000 + Math.floor(Math.random() * 1000);
}

function boilWater() {
    return new Promise((resolve, reject) => {
        setTimeout(() => {
            if (Math.random() < 0.1) {
                reject(new Error("Water boiling failed."));
            } else {
                console.log("Water boiled.");
                resolve();
            }
        }, delay());
    });
}

function brewCoffee() {
    return new Promise((resolve, reject) => {
        setTimeout(() => {
            if (Math.random() < 0.1) {
                reject(new Error("Coffee brewing failed."));
            } else {
                console.log("Coffee brewed.");
                resolve();
            }
        }, delay());
    });
}

function pourCoffee() {
    return new Promise((resolve, reject) => {
        setTimeout(() => {
            if (Math.random() < 0.1) {
                reject(new Error("Pouring coffee failed."));
            } else {
                console.log("Coffee poured into the cup.");
                resolve();
            }
        }, delay());
    });
}

boilWater()
    .then(() => brewCoffee())
    .then(() => pourCoffee())
    .then(() => console.log("Coffee ready for the team!"))
    .catch(error => console.error("Coffee process failed:", error.message));
```